

Analyzing the Impact of Car Features on Price and Profitability

1. PROJECT DESCRIPTION :

The project “Analyzing the Impact of Car Features on Price and Profitability ” is to Investigate the relationship between a car’s features and its popularity: By examining the popularity variable in the dataset, a data analyst could identify which features are most popular among consumers and how they affect a car’s popularity. This could help manufacturers make informed decisions about product development and marketing.

Predicting the price of a car based on its features and market category: By using the various features and market category variables in the dataset, a data analyst could develop a model to predict the price of a car. This could help manufacturers and consumers understand how different features affect the price of a car and make informed decisions about pricing and purchasing.

APPROACH:

I have used the given data set of cars. First of all, I understand the data and then clean up the data. So for cleaning the data what I did firstly I calculated the percentage of blank cells for which I used the following formula
$$=(\text{COUNTBLANK}(A2:A11915)/\text{COUNTA}(\$A\$2:\$A\$11915))*100$$
 and dragged it. Then found that only 4 columns contained some blank cells Engine Fuel Type, Engine HP, Engine Cylinders and Number of Doors. As Engine Fuel Type contains categorical values so I deleted the blank cells and as the other 3 columns are numerical columns, I replaced the blank cells with the median value.

TASKS: ANALYSIS

Task 1A: Create a pivot table that shows the number of car models in each market category and their corresponding popularity scores.

I have created the pivot table as shown below :

Row Labels	Count of Model	Sum of Popularity
Crossover	1110	1715242
Crossover,Diesel	7	6111
Crossover,Exotic,Luxury,High-Performance	1	238
Crossover,Exotic,Luxury,Performance	1	238
Crossover,Factory Tuner,Luxury,High-Performance	26	47410
Crossover,Factory Tuner,Luxury,Performance	5	13037
Crossover,Factory Tuner,Performance	4	840
Crossover,Flex Fuel	64	132720
Crossover,Flex Fuel,Luxury	10	11732
Crossover,Flex Fuel,Luxury,Performance	6	9744
Crossover,Flex Fuel,Performance	6	33942
Crossover,Hatchback	72	120650
Crossover,Hatchback,Factory Tuner,Performance	6	12054
Crossover,Hatchback,Luxury	7	1428
Crossover,Hatchback,Performance	6	12054
Crossover,Hybrid	42	107662
Crossover,Luxury	410	362665
Crossover,Luxury,Diesel	34	73080
Crossover,Luxury,High-Performance	9	9335
Crossover,Luxury,Hybrid	24	15142
Crossover,Luxury,Performance	113	151968
Crossover,Luxury,Performance,Hybrid	2	7832
Crossover,Performance	69	178431
Diesel	84	145396
Diesel,Luxury	51	116025
Exotic,Factory Tuner,High-Performance	21	21974

Exotic,Factory Tuner,Luxury,Performance	3	1560
Exotic,Flex Fuel,Factory Tuner,Luxury,High-Performance	13	6760
Exotic,Flex Fuel,Luxury,High-Performance	11	5720
Exotic,High-Performance	261	331818
Exotic,Luxury	12	1352
Exotic,Luxury,High-Performance	79	36899
Exotic,Luxury,High-Performance,Hybrid	1	204
Exotic,Luxury,Performance	36	7813
Exotic,Performance	10	13910
Factory Tuner,High-Performance	106	205790
Factory Tuner,Luxury	2	1234
Factory Tuner,Luxury,High-Performance	215	458674
Factory Tuner,Luxury,Performance	31	43816
Factory Tuner,Performance	92	156004
Flex Fuel	872	1933488
Flex Fuel,Diesel	16	90512
Flex Fuel,Factory Tuner,Luxury,High-Performance	1	258
Flex Fuel,Hybrid	2	310
Flex Fuel,Luxury	39	29115
Flex Fuel,Luxury,High-Performance	33	29004
Flex Fuel,Luxury,Performance	28	38642
Flex Fuel,Performance	87	146201
Flex Fuel,Performance,Hybrid	2	310
Hatchback	641	845393
Hatchback,Diesel	14	12222
Hatchback,Factory Tuner,High-Performance	13	15667

Hatchback,Factory Tuner,Luxury,Performance	9	7982
Hatchback,Factory Tuner,Performance	22	47499
Hatchback,Flex Fuel	7	39599
Hatchback,Hybrid	72	152730
Hatchback,Luxury	46	63457
Hatchback,Luxury,Hybrid	3	1362
Hatchback,Luxury,Performance	38	59513
Hatchback,Performance	252	261991
High-Performance	199	362468
Hybrid	123	258985
Luxury	855	942772
Luxury,High-Performance	334	557118
Luxury,High-Performance,Hybrid	12	6826
Luxury,Hybrid	52	35029
Luxury,Performance	673	869930
Luxury,Performance,Hybrid	11	25665
N/A	3742	6274920
Performance	601	810673
Performance,Hybrid	1	155
Grand Total	11914	18525212

The quantity of car models in different market categories, along with their respective popularity scores.

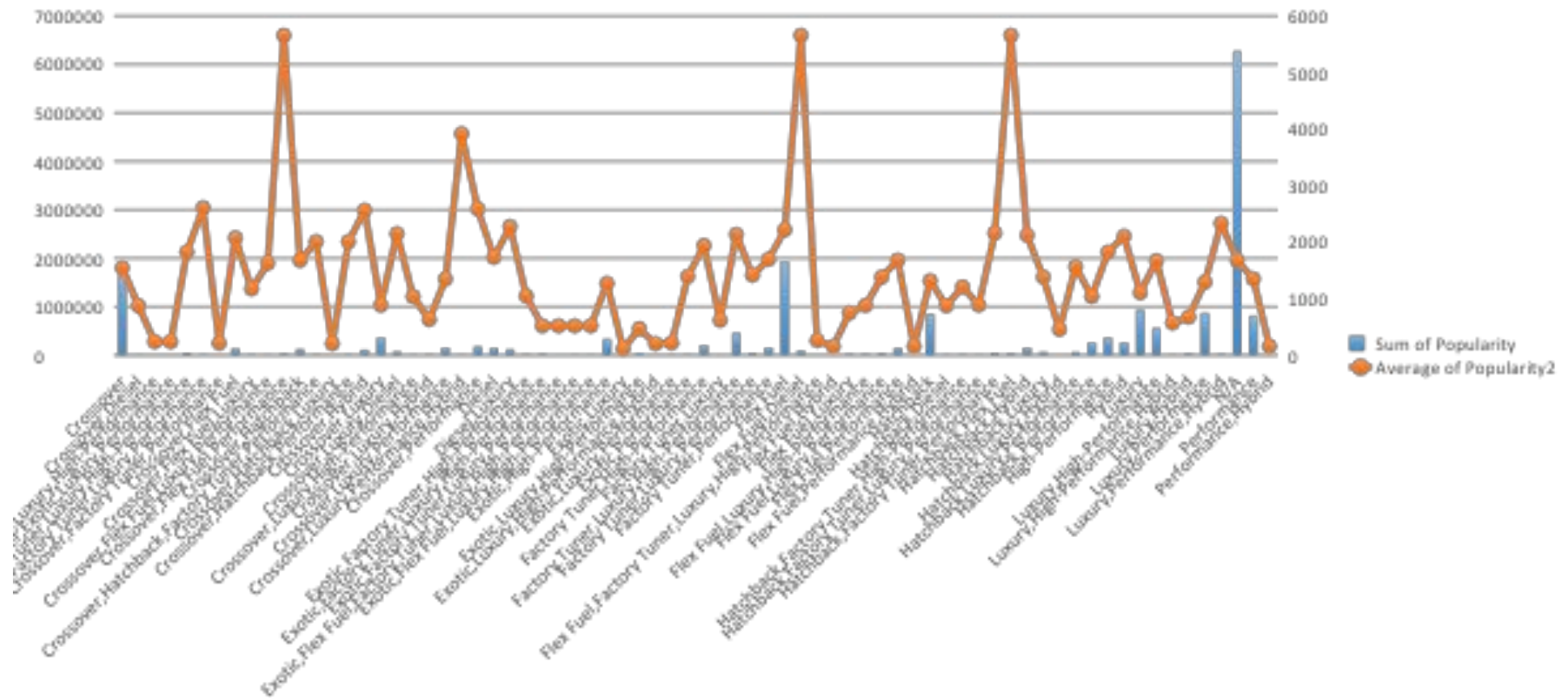
Task 1B: Create a combo chart that visualizes the relationship between market category and popularity

Row Labels	Sum of Popularity	Average of Popularity2
Crossover	1715242	1545.263063
Crossover,Diesel	6111	873
Crossover,Exotic,Luxury,High-Performance	238	238
Crossover,Exotic,Luxury,Performance	238	238
Crossover,Factory Tuner,Luxury,High-Performance	47410	1823.461538
Crossover,Factory Tuner,Luxury,Performance	13037	2607.4
Crossover,Factory Tuner,Performance	840	210
Crossover,Flex Fuel	132720	2073.75
Crossover,Flex Fuel,Luxury	11732	1173.2
Crossover,Flex Fuel,Luxury,Performance	9744	1624
Crossover,Flex Fuel,Performance	33942	5657
Crossover,Hatchback	120650	1675.694444
Crossover,Hatchback,Factory Tuner,Performance	12054	2009
Crossover,Hatchback,Luxury	1428	204
Crossover,Hatchback,Performance	12054	2009
Crossover,Hybrid	107662	2563.380952
Crossover,Luxury	362665	884.5487805
Crossover,Luxury,Diesel	73080	2149.411765
Crossover,Luxury,High-Performance	9335	1037.222222
Crossover,Luxury,Hybrid	15142	630.9166667
Crossover,Luxury,Performance	151968	1344.849558
Crossover,Luxury,Performance,Hybrid	7832	3916
Crossover,Performance	178431	2585.956522
Diesel	145396	1730.904762
Diesel,Luxury	116025	2275
Exotic,Factory Tuner,High-Performance	21974	1046.380952
Exotic,Factory Tuner,Luxury,High-Performance	26912	517.5384615
Exotic,Factory Tuner,Luxury,Performance	1560	520
Exotic,Flex Fuel,Factory Tuner,Luxury,High-Performanc	6760	520
Exotic,Flex Fuel,Luxury,High-Performance	5720	520

Luxury		
Luxury,High-Performance		
Luxury,High-Performance,Hybrid		
Luxury,Hybrid		
Luxury,Performance		
Luxury,Performance,Hybrid		
N/A		
Performance		
Performance,Hybrid		
Grand Total		

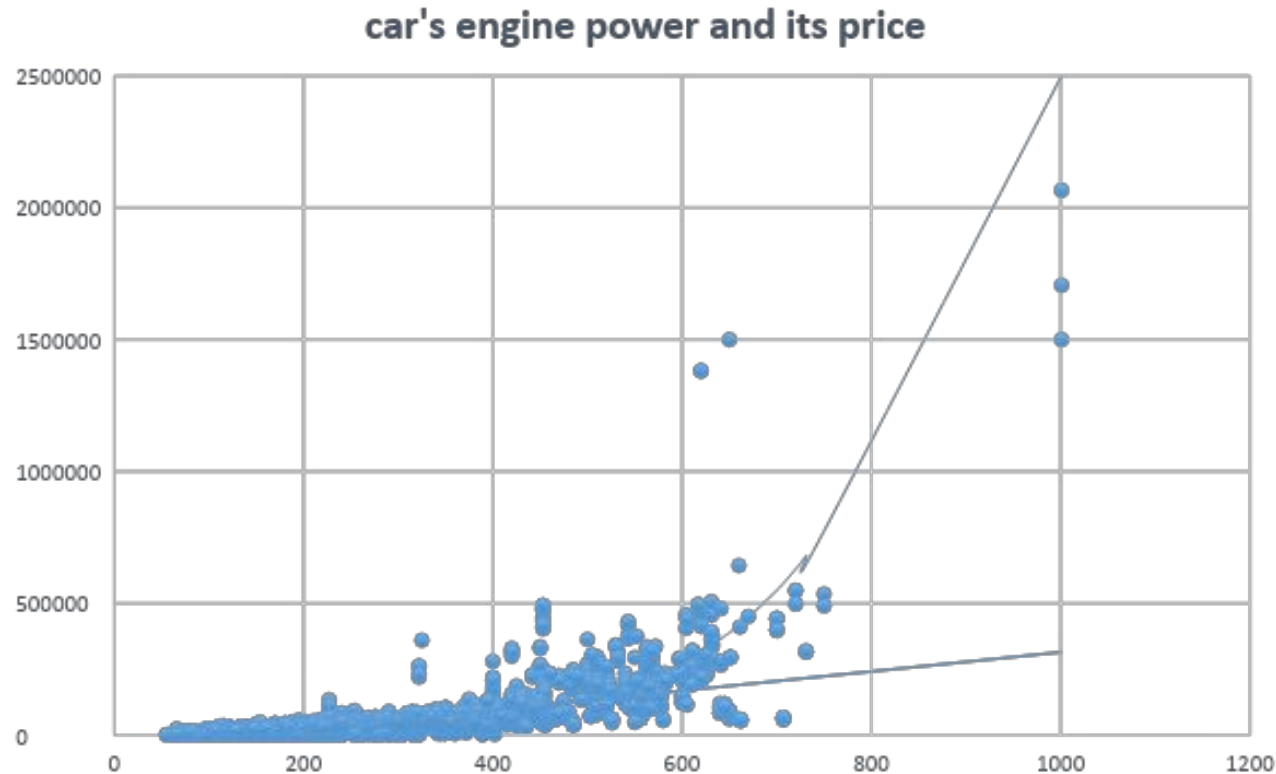
Exotic,High-Performance	331818	1271.333333
Exotic,Luxury	1352	112.6666667
Exotic,Luxury,High-Performance	36899	467.0759494
Exotic,Luxury,High-Performance,Hybrid	204	204
Exotic,Luxury,Performance	7813	217.0277778
Exotic,Performance	13910	1391
Factory Tuner,High-Performance	205790	1941.415094
Factory Tuner,Luxury	1234	617
Factory Tuner,Luxury,High-Performance	458674	2133.367442
Factory Tuner,Luxury,Performance	43816	1413.419355
Factory Tuner,Performance	156004	1695.695652
Flex Fuel	1933488	2217.302752
Flex Fuel,Diesel	90512	5657
Flex Fuel,Factory Tuner,Luxury,High-Performance	258	258
Flex Fuel,Hybrid	310	155
Flex Fuel,Luxury	29115	746.5384615
Flex Fuel,Luxury,High-Performance	29004	878.9090909
Flex Fuel,Luxury,Performance	38642	1380.071429
Flex Fuel,Performance	146201	1680.471264
Flex Fuel,Performance,Hybrid	310	155
Hatchback	845393	1318.865835
Hatchback,Diesel	12222	873
Hatchback,Factory Tuner,High-Performance	15667	1205.153846
Hatchback,Factory Tuner,Luxury,Performance	7982	886.8888889
Hatchback,Factory Tuner,Performance	47499	2159.045455
Hatchback,Flex Fuel	39599	5657
Hatchback,Hybrid	152730	2121.25
Hatchback,Luxury	63457	1379.5
Hatchback,Luxury,Hybrid	1362	454
Hatchback,Luxury,Performance	59513	1566.131579
Hatchback,Performance	261991	1039.646825
High-Performance	362468	1821.447236
Hybrid	258985	2105.569106

	942772	1102.65731
	557118	1668.017964
	6826	568.8333333
	35029	673.6346154
	869930	1292.615156
	25665	2333.181818
	6274920	1676.889364
	810673	1348.873544
	155	155
Grand Total	18525212	1554.911197



Using the pivot table I have plotted the scatter plot as shown above.

Task 2: Create a scatter chart that plots engine power on the x-axis and price on the y-axis. Add a trend line to the chart to visualize the relationship between these variables.



The graph exhibits a positive exponential trend line slope, suggesting a direct correlation between a car's engine power and its price. This implies that vehicles equipped with more powerful engines generally have higher price tags. Overall, If engine power increases price will also increase so we can both have a positive relationship between them.

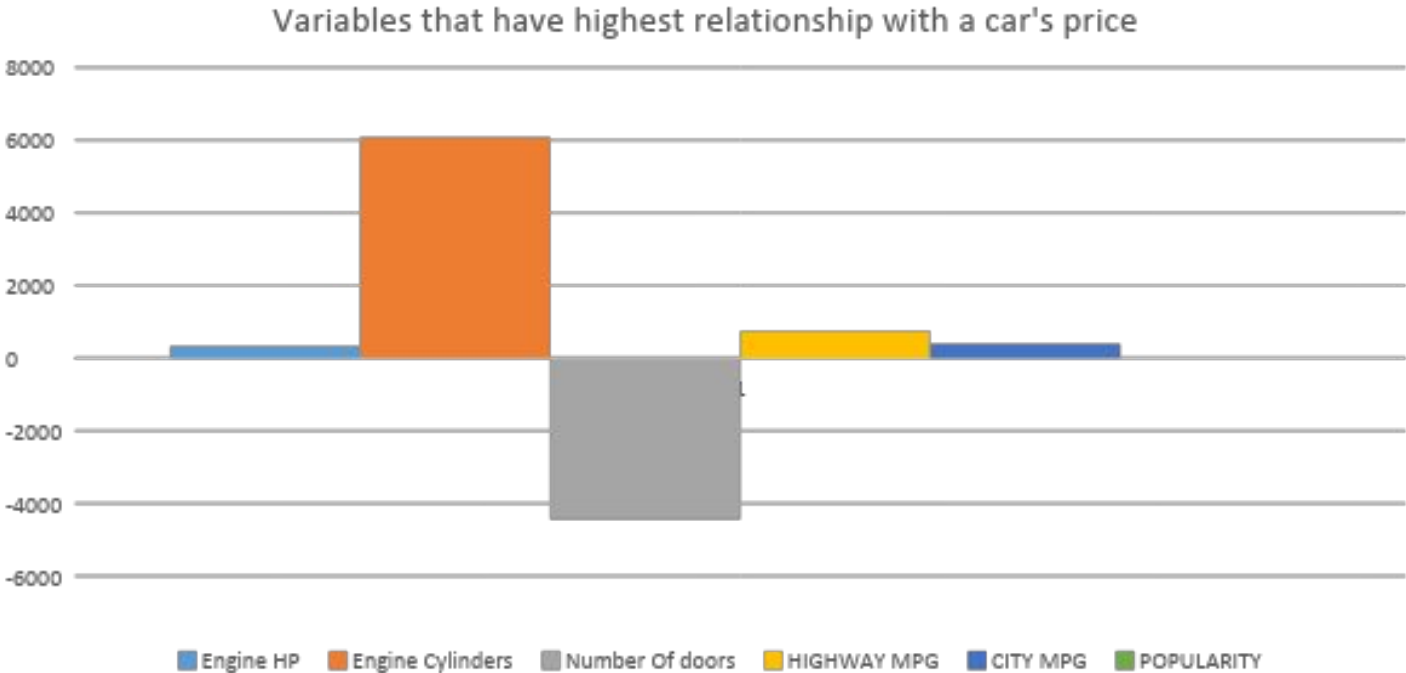
Task 3: Use regression analysis to identify the variables that have the strongest relationship with a car's price. Then create a bar chart that shows the coefficient values for each variable to visualize their relative importance.

SUMMARY OUTPUT	
Regression Statistics	
Multiple R	0.68137613
R Square	0.46427343
Adjusted R Square	0.46400347
Standard Error	44006.9469
Observations	11914

ANOVA					
	df	SS	MS	F	Significance F
Regression	6	1.99837E+13	3.331E+12	1719.815	0
Residual	11907	2.30592E+13	1.937E+09		
Total	11913	4.30429E+13			

	Coefficients
Intercept	-80849.6786
Engine HP	320.260075
Engine Cylinders	6079.46373
Number Of doors	-4448.55262
HIGHWAY MPG	747.434615
CITY MPG	395.951221
POPULARITY	-3.24435623

Based on the regression analysis, the variables of Engine horsepower, Engine cylinders, City MPG, Highway MPG, and MSRP exhibit the most significant relationship with a car's price



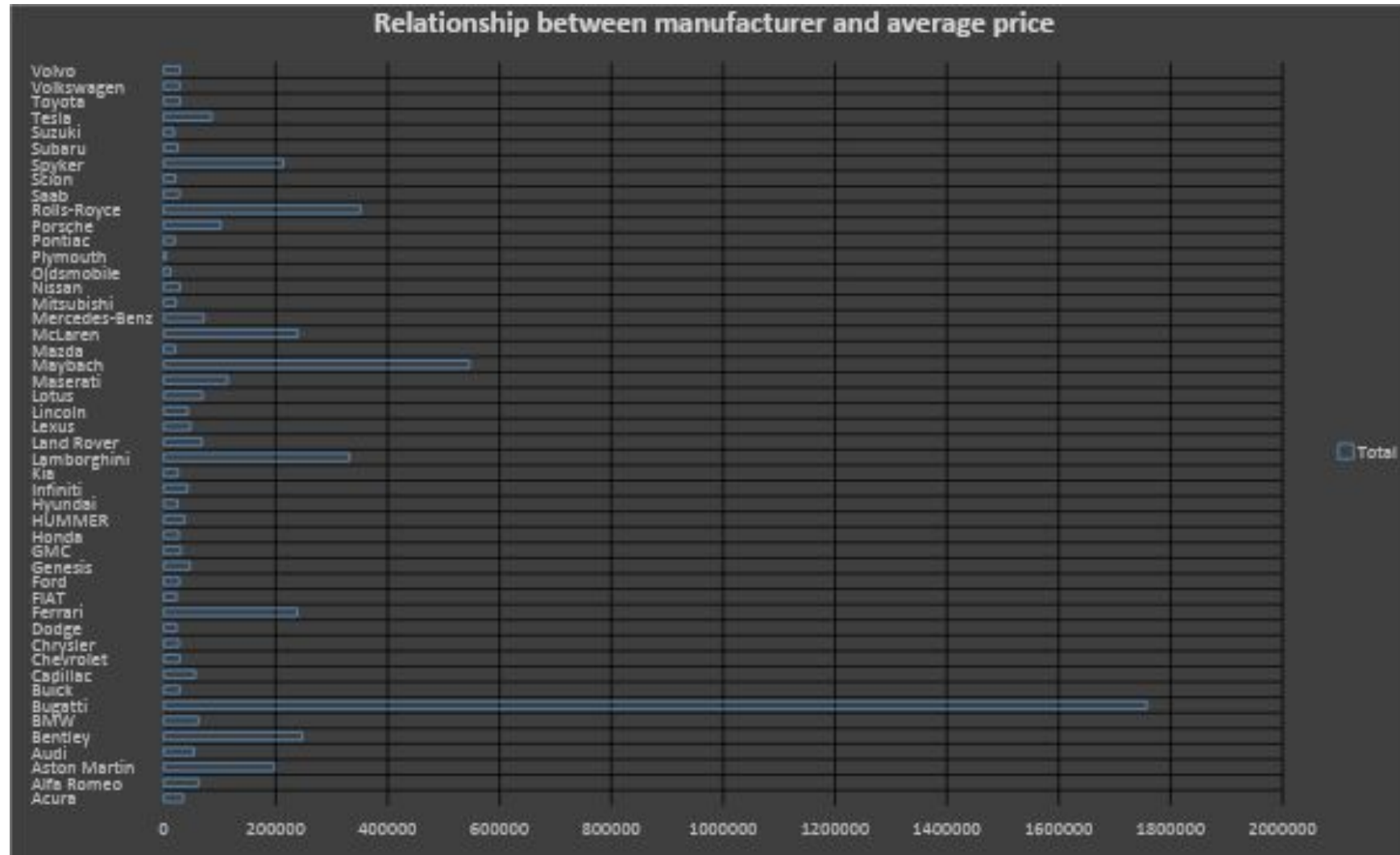
From the bar chart we can conclude that the strongest relationship with price is of Engine Cylinders and the negative relationship is with Number of Doors, which means that as the number of doors in a vehicle increases, the price tends to decrease, and vice versa.

Task 4A: Create a pivot table that shows the average price of cars for each manufacturer.

Row Labels	Average of MSRP		
Acura	34887.5873		
Alfa Romeo	61600	Lotus	69188.27586
Aston Martin	197910.3763	Maserati	114207.7069
Audi	53452.1128	Maybach	546221.875
Bentley	247169.3243	Mazda	20039.38298
BMW	61546.76347	McLaren	239805
Bugatti	1757223.667	Mercedes-Benz	71476.22946
Buick	28206.61224	Mitsubishi	21240.53521
Cadillac	56231.31738	Nissan	28583.4319
Chevrolet	28350.38557	Oldsmobile	11542.54
Chrysler	26722.96257	Plymouth	3122.902439
Dodge	22390.05911	Pontiac	19321.54839
Ferrari	238218.8406	Porsche	101622.3971
FIAT	22670.24194	Rolls-Royce	351130.6452
Ford	27399.26674	Saab	27413.5045
Genesis	46616.66667	Scion	19932.5
GMC	30493.29903	Spyker	213323.3333
Honda	26674.34076	Subaru	24827.50391
HUMMER	36464.41176	Suzuki	17907.20798
Hyundai	24597.0363	Tesla	85255.55556
Infiniti	42394.21212	Toyota	29030.01609
Kia	25310.17316	Volkswagen	28102.38072
Lamborghini	331567.3077	Volvo	28541.16014
Land Rover	67823.21678	Grand Total	40594.73703
Lexus	47549.06931		
Lincoln	42839.82927		

This analysis can offer insights into how prices differ among vehicles produced by each manufacturer, and whether certain manufacturers generally produce cars that are more expensive or less expensive overall.

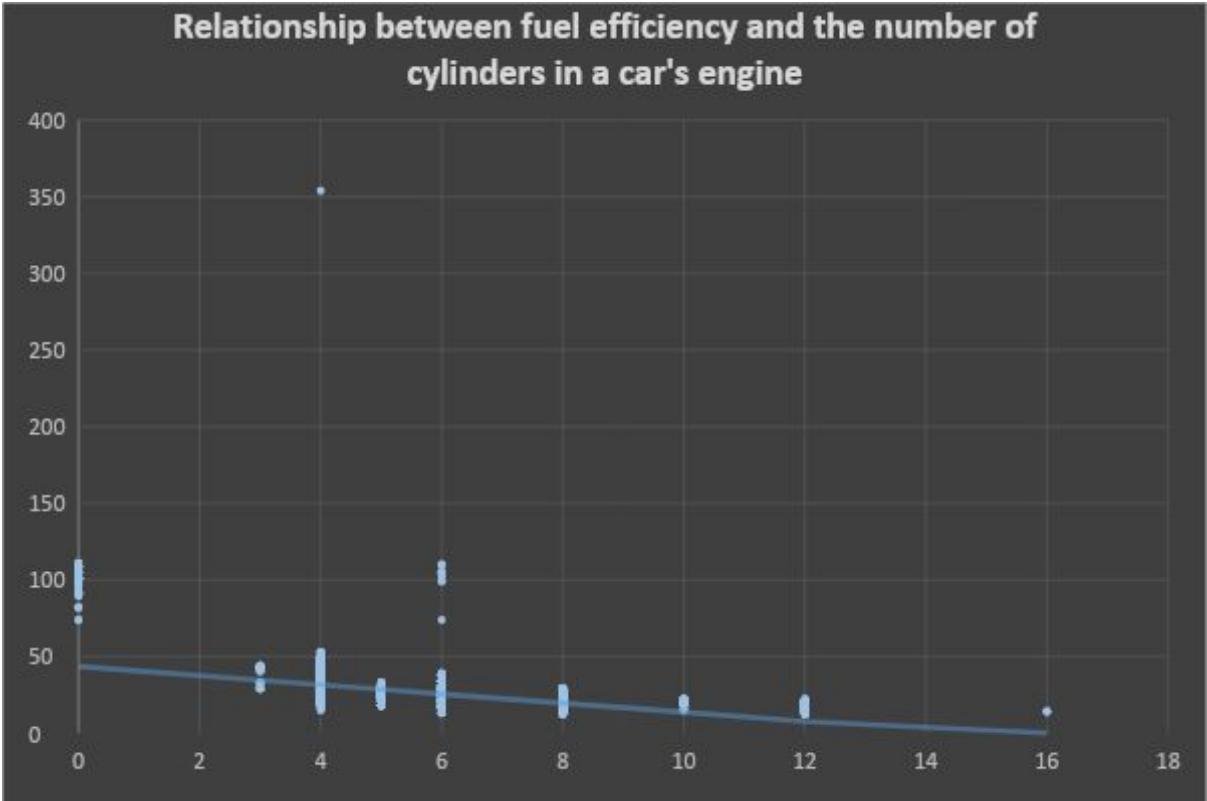
Task 4B: Create a bar chart or a horizontal stacked bar chart that visualizes the relationship between manufacturer and average price.



- Task 5A:** Create a scatter plot with the number of cylinders on the x-axis and highway MPG on the y-axis. Then create a trend line on the scatter plot to visually estimate the slope of the relationship and assess its significance.
- Task 5B:** Calculate the correlation coefficient between the number of cylinders and highway MPG to quantify the strength and direction of the relationship.

	ENGINE CYLINDERS	HIGHWAY MPG
ENGINE CYLINDERS	1	
HIGHWAY MPG	-0.600947254	1

Upon conducting the analysis, it was observed that there exists a negative correlation between the number of cylinders in a car's engine and its highway miles per gallon (MPG). This implies that as the number of cylinders increases, the fuel efficiency of the vehicle tends to decrease. The graph visually depicts this relationship, with the trendline exhibiting a negative slope, indicating a decline in highway MPG as the number of cylinders in the engine increases.

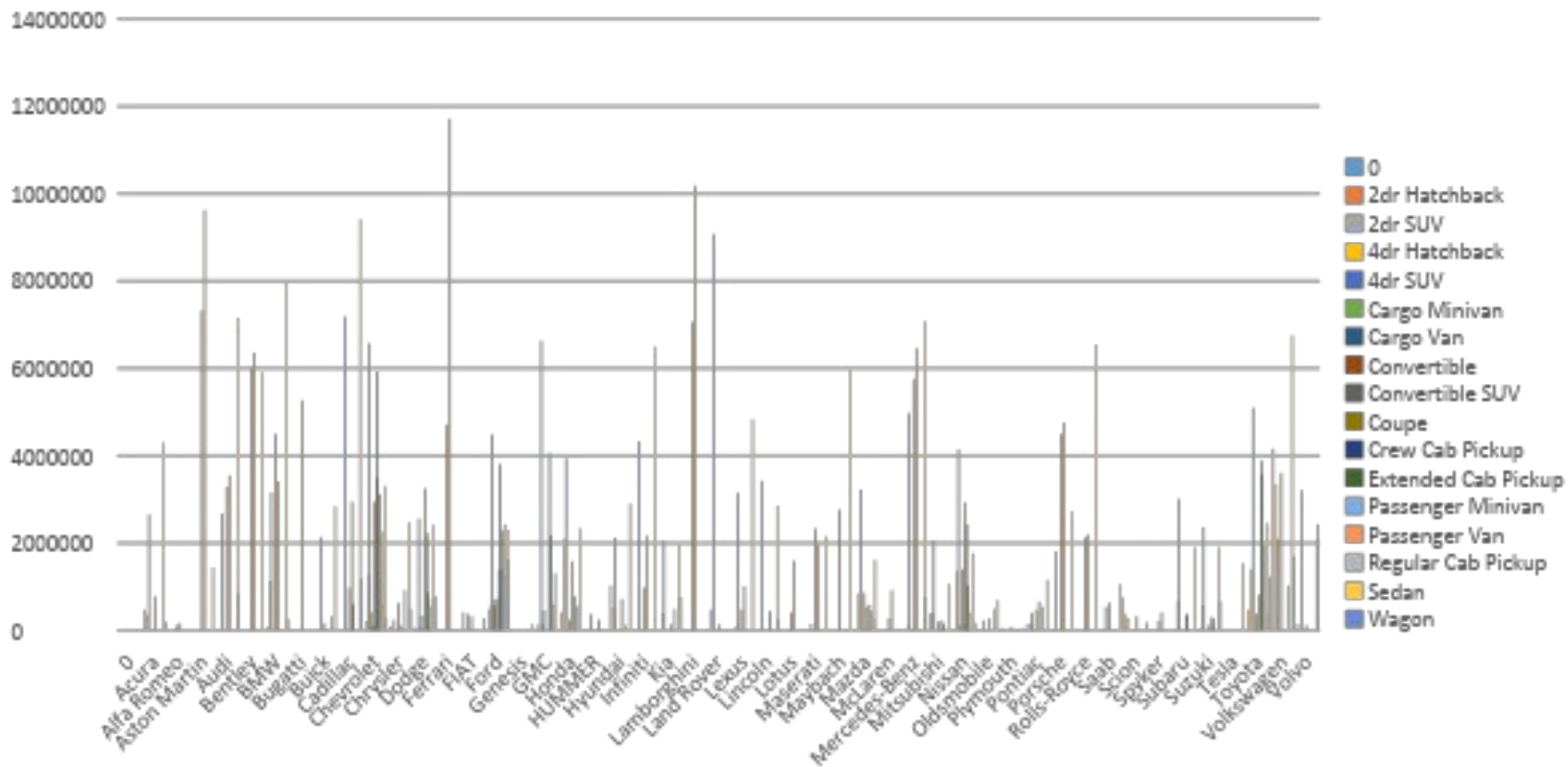


BUILDING THE DASHBOARD

Now for the next portion of the project is to create the Interactive Dashboard.

Task 1: How does the distribution of car prices vary by brand and body style?

Row Label	0	2dr Hatchb	2dr SUV	4dr Hatchb	4dr SUV	Cargo Mini	Cargo V	Convertil	Convertible	Coupe	Crew Cab	Pic	Extended Cab	P	Passenger Mii	Passenger	Regular Cab	P Sedan	Vagon	Grand Total
0	0																			0
Acura		480917		357440	2663505					793748								4294702	201360	8791672
Alfa Romeo								129800		178200										308000
Aston Martin								7321655		9635275								1448735		18405665
Audi		4000			2674900			3291405		3556290								7158348	847350	17532293
Bentley								6012870		6356760								5920900		18290530
BMW		80097		1144950	3160950			4502671		3419051								7989300	259600	20556619
Bugatti										5271671										5271671
Buick					2141770			179325		18534					330065			2850590	8212	5528496
Cadillac					7182555			985607		2953574	599150							9418847	1184100	22323833
Chevrolet		8000	213310	1287260	6569568	420150	78688	2953245	106300	3504525	5927617		3117951		1178515	607670	2260032	3303977	300675	31837483
Chrysler		98805			250545			630105		114510					922295			2479859	501075	4997194
Dodge		48000	44000	18000	2572405	60520	338497	12000		3264627	2235775		864172		557425	70708	719408	2417585	793055	14016177
Ferrari								4723811		1.2E+07										16437100
FIAT		420715			369305			327965											287570	1405555
Ford		36000	479873	567615	4482771	702400	566351	730007		1398144	3812353		2285584		1411605	2431898	1299240	2299348	1635565	24138754
Genesis																		139850		139850
GMC			144319		6641919	142750	468085				4062482		2183866		150630	603670	1306328			15704049
Honda		413200		2088520	3953209			252135		1588705	787720				553185			2340105		11976779
HUMMER					377490						242405									619895
Hyundai		1038050		528880	2128890					724070					133075			2899937		7452902
Infiniti					4340200			980050		2175750								6494090		13990090
Kia				406960	2049645					142630						494650		1980360	772405	5846650
Lamborghini								7064450		1E+07										17241500
Land Rover			476394		9076595				145731											9698720
Lexus				94700	3152974			472065		1016472								4837596	31105	9604912
Lincoln					3422570					25342	453260							2854855	269705	7025732
Lotus								413260		1593200										2006460
Maserati					155000			2342963		1972284								2153800		6624047
Maybach								2762750										5976800		8739550
Mazda		22000	24000	853180	3222525			870505		543879			580033		443130		265486	1618571	33350	8476659
McLaren								280225		918800										1199025
Mercedes-Benz				122800	4974610	28950		5753964		6473107					32500			7080243	764935	25231109
Mitsubishi		394868		407835	2066505	2000		209893			240210		134360		2000		8000	1058563		4524234
Nissan		14683		1347320	4149630	128620		1406552	131075	2943632	2422300		1026379		413320		21914	1769130	175000	15949555
Oldsmobile					238150			2000		286015					492055			691161	22000	1731381
Plymouth		42000		16000				85631		14000					33688			46759	18000	256078
Pontiac		163505		162975	401550			473481		667715					541192			1160535	22855	3593808
Porsche		28827			1815200			4504586		4758533								2713500		13820646
Rolls-Royce								2141365		2204675								6539010		10885050
Saab		14000		36586	541905			632628										1066500	751280	3042899
Scion		366325		282470						330210								32500	184445	1195950
Spyker								219990		419980										639970
Subaru		12000		678060	3020230					356476	365975							1913100	10000	6355841
Suzuki		46496	14000	584387	2362141				122194		304131		259659					1906715	685707	6285430
Tesla																		1534600		1534600
Toyota		473750		1397750	5106450			386668		811995	3893760		3558504		1956518		373446	2459596	1237955	21656392
Volkswagen		4171275		3355760	2084955			3612631		8000					1038130			6760050	1704025	22734826
Volvo		157550			3219000			121600		6000								2086945	2428971	8020066
Grand Total	0	8535063	1E+06	15739448	1E+08	1485390	1E+06	6.7E+07	505300	9E+07	25347138		14010508		10683978	3713946	6253854	1.2E+08	2E+07	483645697



This analysis offers valuable insights into the variations in car prices based on brand and body style. Such insights can prove beneficial for manufacturers in optimizing their pricing strategies and enhancing profitability. Additionally, the utilization of slicers enables a deeper exploration of the data, allowing for a more detailed examination of specific details and patterns. The following brand Chevrolet, Mercedes-Benz and Ford have the highest sum MSRP, similarly the vehicle styles sedan, coupe and passenger minivan have the highest sum MSRP. Hence, Chevrolet has the highest price distribution

Make

0

Acura

Alfa Romeo

Aston Martin

Audi

Bentley

BMW

Bugatti

Vehicle Style

0

2dr Hatchback

2dr SUV

4dr Hatchback

4dr SUV

Cargo Minivan

Cargo Van

Convertible

MSRP

0

2000

2002

2003

2008

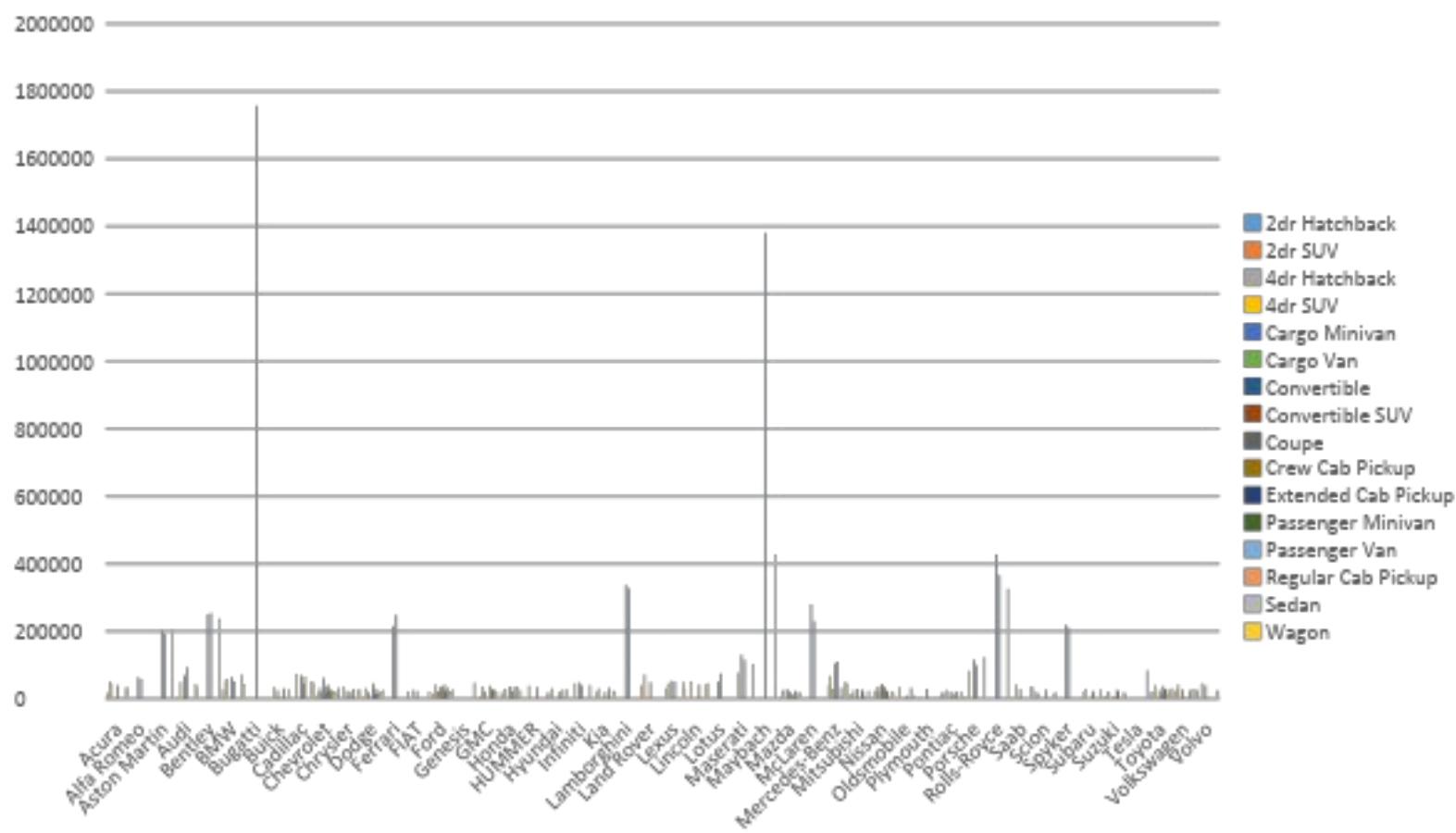
2012

2013

2019

Task 2: Which car brands have the highest and lowest average MSRPs, and how does this vary by body style?

Average of MSRP																		
Row Labels	2dr Hatchback	2dr SUV	4dr Hatchback	4dr SUV	Cargo Minivan	Cargo Van	Convertible	Convertible S Coupe	Crew Cab	Pickup	Extended Cab	Pickup	Passenger Minivan	Passenger 'Regular	Cab	Pickup Sedan	Wagon	Grand Total
Acura	17175.60714		51062.8571	42959.7581				39687.4								33292.2636	33560	34887.5873
Alfa Romeo							64900	59400										61600
Aston Martin							203379.306	192705.5								206962.143		197910.376
Audi	2000			48634.5455			70029.8936	93586.5789								44461.7888	33894	53452.1128
Bentley							250536.25	254270.4								236836		247169.324
BMW	26699		54521.4286	58536.1111			63417.9014	51803.803								70701.7699	43266.6667	61546.7635
Bugatti								1757223.67										1757223.67
Buick				33996.3492			25617.8571	2059.33333					30005.90909			27946.9608	2053	28206.6122
Cadillac				72551.0606			70400.5	45439.6	66572.22222							50912.6865	47364	56231.3174
Chevrolet	2000	8887.91667	18930.2941	32046.6732	20007.1429	7153.45455	62835	17716.66667	38939.1667	39255.74172	24170.16279	24552.39583	24306.8	19824.84211	20521.5963	15825	28350.3856	
Chrysler	32935			35792.1429			24234.8077	19085				29751.45161			26103.7789	26372.3684	26722.9626	
Dodge	2000	2000	2000	30992.8313	20173.3333	12536.9259	2000	45980.662	31052.43056	13938.25806	25337.5	14141.6	9342.961039	21780.045	24782.9688	22390.0591	238218.841	
Ferrari							214718.682	249218.915										238218.841
FIAT	21035.75			24620.3333			23426.0714										22120.7692	22670.2419
Ford	2000	13710.6571	19572.931	41507.1389	21284.8485	17698.4688	34762.2381	34101.0732	41438.61957	23808.16667	23526.75	32425.3067	17797.80822	21290.2593	27259.4167	27399.2667	27399.2667	
Genesis																46616.6667		46616.6667
GMC		5550.73077		36695.6851	23791.6667	18723.4			39062.32692	26632.5122	25105	26246.5217	21069.80645				30493.299	
Honda	17216.66667		26106.5	28855.5401			36019.2857	21763.0822	34248.69565		36879				26001.1667		26674.3408	
HUMMER				37749					34629.28571									36464.4118
Hyundai	18536.60714		17629.3333	30412.7143				20687.7143				26615			27102.215		24597.0363	
Infiniti				45686.3158			46669.0476	40291.6667							40588.0625		42394.2121	
Kia			19379.0476	31533				20375.7143				32976.66667			23298.3529	20326.4474	25310.1732	
Lamborghini							336402.381	328291.935										331567.308
Land Rover		39699.5		70910.8984				48577										67823.2168
Lexus			31566.6667	45042.4857			52451.6667	50823.6							48864.6061	31105	47549.0693	
Lincoln				50331.9118				2111.83333	41205.45455						42609.7761	44950.8333	42839.8293	
Lotus							51657.5	75866.6667										69188.2759
Maserati				77500			130164.611	116016.706							102561.905		114207.707	
Maybach				1381375											426914.286		546221.875	
Mazda	2000	2000	20809.2683	27080.042			28080.8065	20143.6667		11600.66	23322.63158		9154.689655	19738.6707	16675	20039.383		
McLaren				280225				229700										239805
Mercedes-Benz			40933.3333	68145.3425	28950		104617.527	109713.678				32500			49168.3542	44996.1765	71476.2295	
Mitsubishi	13162.26667		13155.9677	26158.2911	2000		29984.7143		26690	19194.28571		2000			24058.25		21240.5352	
Nissan	2097.571429		24059.2857	34294.4628	21436.6667		39070.8889	43691.66667	34228.2791	32733.78378	20527.58	22962.22222		2191.4	21841.1111	17500	28583.4319	
Oldsmobile				34021.4286			2000	9226.29032				32803.66667			8131.30588	2000	11542.54	
Plymouth	2000		2000				28543.6667	2000				2105.5			2597.72222	2000	3122.90244	
Pontiac	18167.22222		18108.3333	25096.875			22546.7143	15528.2558				20815.07692			20009.2241	5713.75	19321.5484	
Porsche	5765.4			82509.0909			115502.205	99136.1042							123340.909		101622.397	
Rolls-Royce							428273	367445.833							326950.5		351130.645	
Saab	2000		2032.55556	41685			28755.8182								36775.8621	34149.0909	27413.5045	
Scion	20351.38889		15692.7778					27517.5							16250	18444.5	19932.5	
Spyker							219990	209990										213323.333
Subaru	2000		21189.375	29322.6214				15498.9565	24398.33333						26570.8333	2000	24827.5039	
Suzuki	6642.285714	2000	16696.7714	21090.5446				7187.882353	27648.27273	21638.25					18159.1905	15237.9333	17907.208	
Tesla															85255.5556		85255.5556	
Toyota	18950		22186.5079	40851.6			25777.8667	15615.2885	37803.49515	26359.28889	29201.76119		16236.78261	24844.404	31742.4359	29030.0161		
Volkswagen	24251.59884		27964.6667	41699.1			27789.4692	2000			25320.2439			29911.7257	25818.5606	28102.3807		
Volvo	26258.33333			45338.0282			40533.3333	2000						20869.45	24785.4184	28541.1601		
Grand Total	16867.7134	10115.19	22420.87	40421.87	20920.99	15280.22	84224.28	17424.1379	76248.32	37220.467	22488.77689	25621.05036	29015.2	15953.70918	39270.69	25557.94	40594.74	



Make
Acura
Alfa Romeo
Aston Martin
Audi
Bentley
BMW
Bugatti
Buick

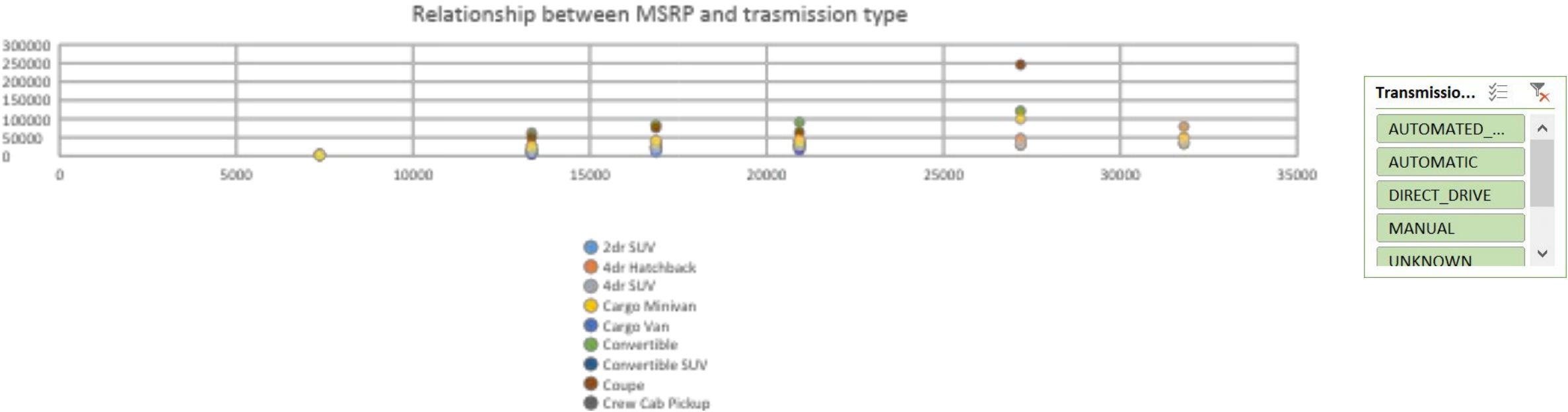
Vehicle Style
2dr Hatchback
2dr SUV
4dr Hatchback
4dr SUV
Cargo Minivan
Cargo Van
Convertible
Convertible SUV

MSRP
2000
2002
2003
2008
2012

During this analysis, it was observed that certain brands exhibit significantly higher or lower average Manufacturer's Suggested Retail Price (MSRP) compared to others. Luxury brands like Bugatti, Maybach, and Rolls Royce tend to have higher average MSRP values when compared to brands such as BMW, Toyota, and Audi. Additionally, specific vehicle styles such as Sedan, 4Dr SUV, and Coupe tend to have higher average MSRP values compared to other styles. This highlights the price differentials based on brand and vehicle style, emphasizing the varying price ranges within the automotive market. The results of this analysis demonstrate the evolution of fuel efficiency over time across different body styles. Hence, Bugatti has the highest MSRP and

Task 3: How do the different features such as transmission type affect the MSRP, and how does this vary by body style?

Average of MSRP	Column Labels																
Row Labels	2dr Hatchback	2dr SUV	4dr Hatchback	4dr SUV	Cargo Minivan	Cargo Van	Convertible	Convertible SUV	Coupe	Crew Cab Pickup	Extended Cab Pickup	Passenger Minivan	Passenger Van	Regular Cab	Pi Sedan	Wagon	Grand Total
AUTOMATED_MANUAL	27180.96491		29249.0741	40451.1538			121256.644		245977.425						47498.7	31985.2778	99508.3706
AUTOMATIC	20926.464	18615.2045	23833.679	41535.6065	20920.9859	15280.2211	90637.3869	38925.5	63371.8108	37744.07154	30637.34973	26412.68159	29015.2031	28536.8239	43769.1	27613.1917	41110.3317
DIRECT_DRIVE	31800		32799.7297	49800											79512.3	34250	47351.25
MANUAL	13353.65831	6303.81111	17594.4131	15426.4623			62357.7563	9233.142857	50484.3724	28360.52632	10884.19455	4405.333333		7557.77333	17119.2	17844.1397	26663.6443
UNKNOWN	7361.5	2371					5783.5		2000					2000	2000		3040.73684
Grand Total	16867.71344	10115.1884	22420.8661	40421.8718	20920.9859	15280.2211	84224.285	17424.13793	76248.322	37220.46696	22488.77689	25621.05036	29015.2031	15953.7092	39270.7	25557.9392	40594.737

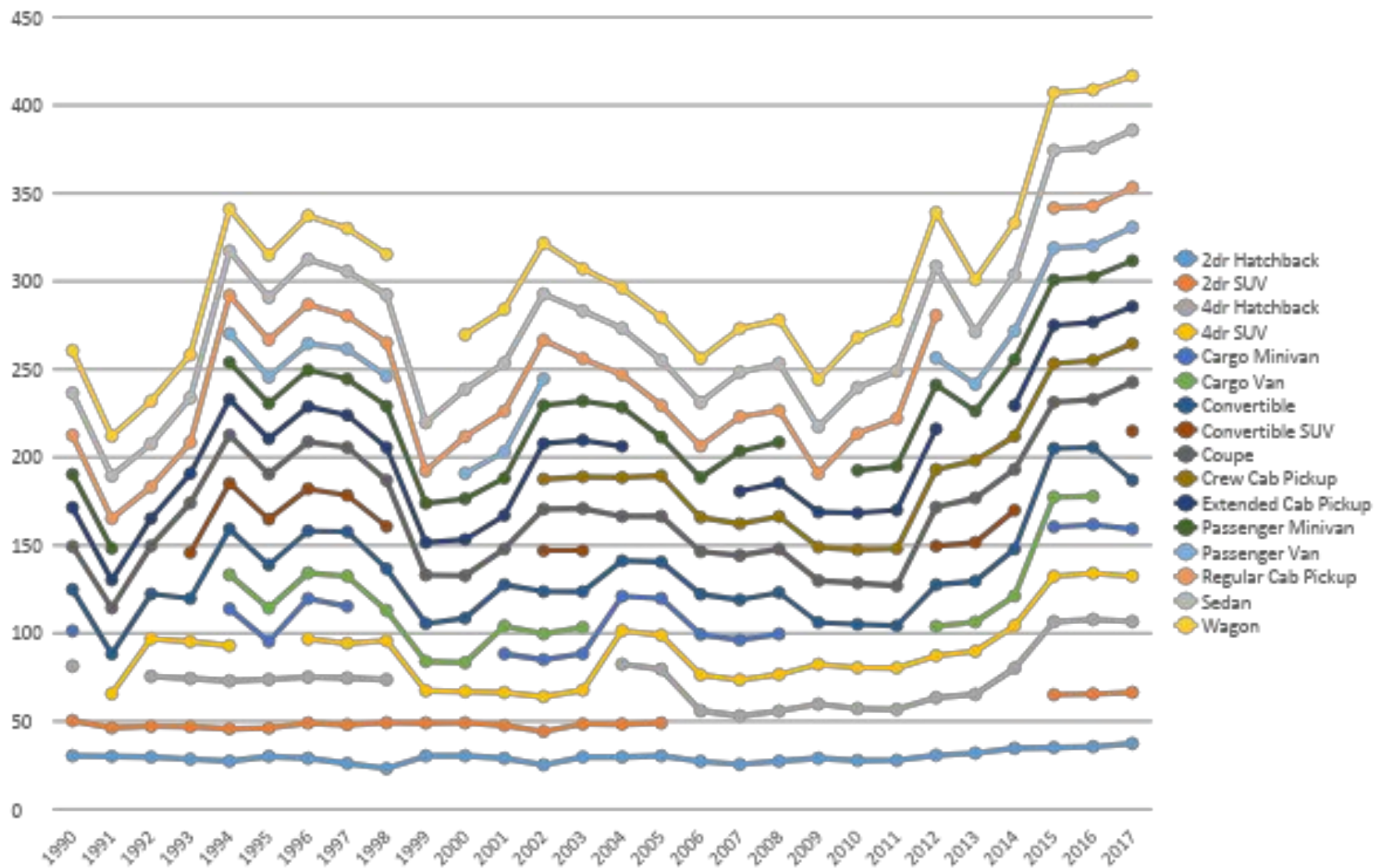


Automatic_manual is the most expensive category and the most popular also.

Task 4: How does the fuel efficiency of cars vary across different body styles and model years?

Average of highway		Column Lab																		
Row Labels	T	2dr Hatchback	2dr SUV	4dr Hatchb.	4dr SUV	Cargo Miniv	Cargo Van	Convertible	Convertible S	Coupe	Crew Cab	Pic	Extended Cab	Pi	Passenger Min	Passenger 'Regular	Cab Pi	Sedan	Wagon	Grand Total
1990		30.4	20	31		20		23.5		24.5				22	18.85714286		22.23076923	24	24.1333333	23.0731707
1991		30.06666667	16.25		19.33333333			22.625		26.15789				15.83333333	18		16.95238095	24.2195	22.5714286	22.1513158
1992		29.6969697	17.4705882	28.375	21.33333333			25.5		26.28571				15.6			17.88235294	24.5208	22.666667	24.0508475
1993		28.53333333	18.4736842	27.3	21			24.4615385	26	28.25926				16.71428571			17.64705882	25.3276	24.4615385	24.2105263
1994		27.35	18.4285714	27.1428571	20	21	19.33333333	26	26	27.29167				20.28571429	21	16.4	21.66666667	25.2273	23.8333333	23.8650307
1995		30.14285714	16	27.6666667		21.5	19	24.5	26	25.67742				20	20.08333333	15	21.2	24.0645	24.1	23.2296296
1996		29	20	26.125	21.6	23	14.55555556	23.8	24	26.72727				20	20.77777778	15	22.2	25.7273	24.6666667	23.7251908
1997		26.11111111	22	26.5	19.7	21	17.125	25.2857143	20.66666667	27.2069				18.35714286	20.55555556	17	18.78571429	25.3182	24.4	22.3085714
1998		23.2	26	24.5	22.1111111			17.2	23.6666667	24	26.26667			18.625	23.4	17	19.15151515	27.12	23	21.8506494
1999		30.33333333	18.75		18.3		16.6666667	21.5		27.55556				18.42307692	22.33333333		18.42857143	27.4054		22.975
2000		30.41666667	18.75		17.73333333		16.4	25.2857143	24.16667					20.5	23.16666667	14.5	20.83333333	26.8444	31	24.0423729
2001		29	18.6666667		18.7272727	22	15.8	23.4375		20.29412				19	21.2	15	23	27.3774	30.625	23.7083333
2002		25.25	19		19.7941176	21	14.6	24.0714286	23.28571429	23.6		17		20.22222222	21.6875	15	22.06666667	26.14	28.8888889	22.7658537
2003		29.75	18.75		19.2285714	20.6666667	15	20.2307692	23.4	23.87879			18	20.77777778	22.2972973		24.08333333	27.0577	24	22.7352941
2004		29.71428571	18.75	34	19.0408163	19.6		20.1		25.26667		22		17.75			18.46153846	26.3623	22.8	23.1489362
2005		30.33333333	18.6666667		30.6	19.33333333	20.8571429	20.7272727		26		23			21.88888889		18	25.7541	24.2777778	23.5868545
2006		27.25			28.75	20.3555556	23	22.8571429		24.25926	19.38461538				22.5		18	24.75	25	23.4243902
2007		25.53846154		27.5384615	20.5084746	22.6666667		22.76		25.2	18.03333333		18.38983051		22.77777778		19.57692308	25.3077	24.8	21.9391304
2008		27.31818182		28.5	20.7941176	23		23.5111111		24.78947	18.45652174			19.2	23		18	26.72	24.7142857	22.9885387
2009		29		30.8	22.5913978			23.7619048		23.74359	19.05405405		19.875				21.85714286	26.6329	26.8484848	23.8601583
2010		27.76923077		29.5	23.1929825			24.6190476		23.52174	18.94871795		20.78947368		24.2		21	26.0606	28.4782609	24.1442953
2011		27.83333333		28.9310345	23.58333333			23.9444444		22.67857		21.1		21.9	25		27	26.9744	28.7333333	25.1754386
2012		30.73913043		32.7037037	23.8444444		16.6666667	23.5769231		22.08108	21.43333333		23.0625		25	15.3333333	24.125	27.9333	30.5526316	26.4728682
2013		31.94736842		33.3731343	24.4736842		16.6666667	23.1818182	22	25.25	21.31818182				28	15.3333333		29.75	29.5952381	27.7540984
2014		34.7037037		45.4680851	24.2231405		16.8571429	26.6477273	22	23.19231	18.86666667		17.4		26	16.375		32.0342	29.375	28.0458404
2015		35.12359551	30	41.5172414	25.770073	28	17	27.625		26.22396	22.01481481		21.65934066		25.78181818	18.1428571	22.74285714	32.7624	32.7462687	29
2016		35.56451613	30	42.28	26.2453469	27.6923077	16	27.8769231		27.0765	22.25714286		21.79775281		25.63636364	17.7142857	22.52941176	33.1016	32.9836066	29.2021326
2017		37.4375	29	40.2941176	25.75625	26.6666667		27.8026316	28	27.73469	21.84782609		21.01298701		26.12820513	19	22.52941176	32.597	30.8648649	28.5773381
Grand Total		31.3754941	19.11594	37.56125	24.49277	24.47887	16.61053	25.78815	23.7241379	25.714	21.0572687		20.1364366		23.56115108	17.17969	20.61989796	30.24	28.35473	26.637485

During the analysis, it was observed that body styles such as 4Dr hatchback, 2Dr hatchback, and sedan tend to exhibit higher overall fuel efficiency. This suggests that these particular body styles have shown a greater focus on fuel efficiency improvements compared to other styles.



Year

1990

1991

1992

1993

1994

1995

1996

1997

Vehicle Style

2dr Hatchback

2dr SUV

4dr Hatchback

4dr SUV

Cargo Minivan

Cargo Van

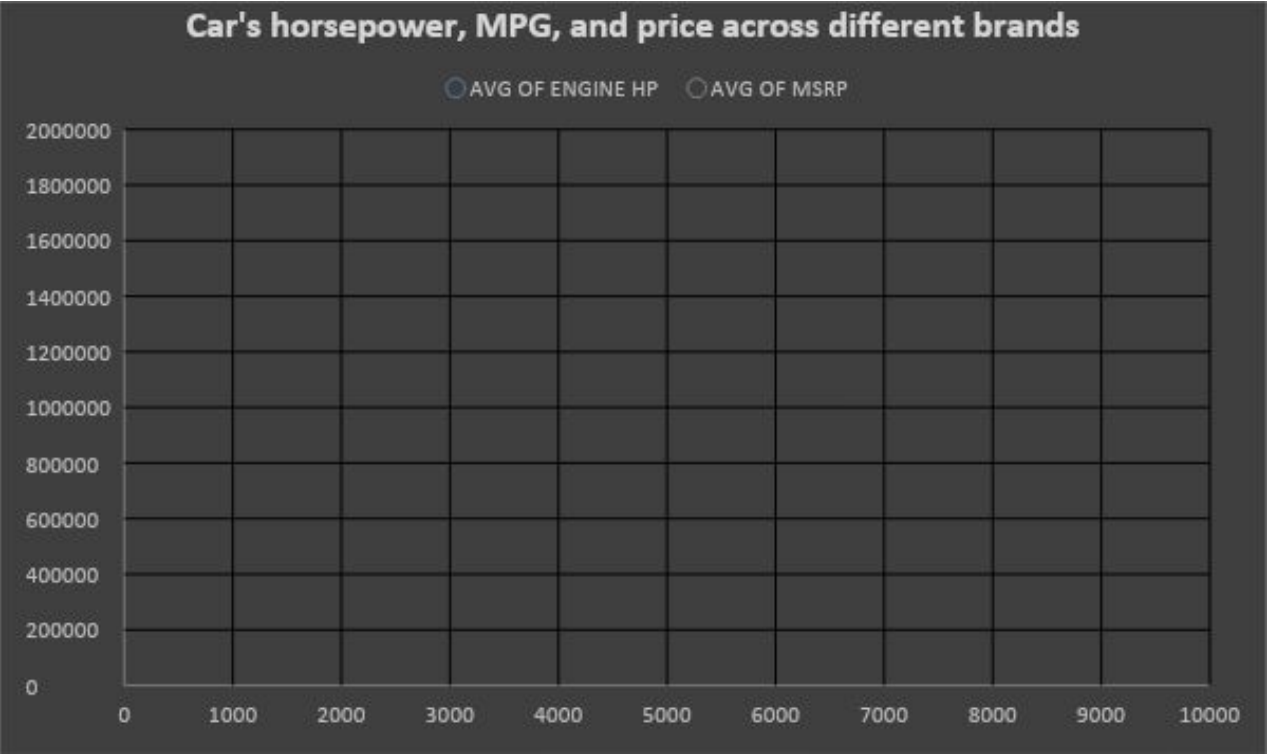
Convertible

Convertible SUV

Over the year fuel efficiency is increasing at a slow speed.

Task 5: How do the car’s horsepower, MPG, and price vary across different Brands?

Row Labels	Average of Engine HP	Average of highway MPG	Average of MSRP
2	100	34.5	15821.66667
3	165.9433962	38.47169811	22195
5	157	28	22183.63636
6	184	36.73333333	24761.33333
57	586.2857143	16	401357.1429
62	586.2857143	16	452471.4286
80	126.3333333	23.66666667	2000
86	202.5	30	26615
90	172	23.22222222	2000
100	172	22.26666667	2000
200	210.0344828	31.20689655	21496.2069
240	114	25.5	2000
300	294.6666667	28.83333333	38134.33333
323	82	32.8	2000
360	401.9230769	15	161978.5385
500	116.0714286	36.57142857	20449.82143
550	485	12	227170
599	621	15	334449.2
626	148.6363636	26.09090909	13786.81818
740	133.875	24.1875	2000
760	156	19.66666667	2000
780	166	19.5	2000
850	202.9230769	24.76923077	2003.5
900	173.2608696	24.82608696	2025.130435
911	426.3793103	26.15517241	123763.7931



The analysis reveals that certain brands exhibit higher average horsepower, while others have lower average horsepower. This trend holds true for average miles per gallon (MPG) and manufacturer's suggested retail price (MSRP) as well. In other words, there are variations in average horsepower, MPG, and MSRP across different brands, indicating disparities in performance, fuel efficiency, and pricing within the automotive market. If engine hp increases highway mpg will decrease and the price will also increase.

3. Tech-Stack Used:

- For this project I have used Excel 2016 which is a spreadsheet application in the Microsoft Office 2016 suite. It is used for tasks like data entry, analysis, and visualization. Excel 2016 provides tools for creating tables, charts, and performing calculations. It's a widely used program for managing and analyzing data.

https://docs.google.com/spreadsheets/d/16g-nc4yCMkoycbp8IMcusnpH_3xD8a4U/edit?usp=sharing&ouid=112612761911126686527&rtpof=true&sd=true

Above is the hyperlink of the excel file that I have worked with.

4. Insights:

The insights and knowledge I have gained while working on this project are as follows:

- Excel formulas automate calculations, saving time and reducing errors compared to manual calculations.
- Excel formulas enable data analysis, summarization, and extraction of insights from large datasets.
- Excel charts provide a visual representation of data, making it easier to understand and interpret complex information.
- Excel charts help in identifying trends, outliers, and correlations within data, which may not be immediately apparent in raw numbers.

5. Result:

Data Analysis in Excel for Car Pricing and Profitability:

1. Market Analysis:

- Created a pivot table to showcase car models' popularity scores in each market category.

Dashboard Creation:

1. ****Price Distribution by Brand and Body Style:****

- Employed Excel's tools to create visualizations (histograms/box plots) showcasing car price distribution across brands and body styles.

2. ****Average MSRPs by Brand and Body Style:****

- Utilized Excel to generate bar charts illustrating the highest and lowest average MSRPs, allowing for brand and body style comparisons.

3. ****Feature Impact on MSRP:****

- Employed Excel visualizations (box plots/bar charts) to highlight the impact of features, such as transmission type, on car prices across different body styles.

4. ****Fuel Efficiency Variation:****

- Developed line charts or heatmaps in Excel to illustrate variations in fuel efficiency across body styles and model years.

5. ****Brand-wise Variation in Horsepower, MPG, and Price:****

- Excel scatter plots and grouped bar charts were utilized to depict variations in horsepower, MPG, and price across different car brands.

Project Impact:

Through the Excel-centric approach, the project successfully delivered actionable insights for optimizing pricing, enhancing product development strategies, and maximizing profitability in the ever-evolving automotive industry.

2. Visualizations:

- Developed combo charts to visually represent the relationship between market categories and popularity.

3. Scatter Chart with Trend line:

- Utilized Excel to create a scatter chart with engine power vs. price, adding a trend line to illustrate their relationship.

4. Regression Analysis:

- Conducted regression analysis in Excel to identify variables impacting car prices. Represented results in a bar chart.

5. Manufacturer Profiling:

- Developed a pivot table to display average car prices for each manufacturer.

6. Manufacturer vs. Price Visualization:

- Created a bar chart in Excel to visualize the relationship between manufacturers and average car prices.

7. Scatter Plot and Correlation:

- Produced a scatter plot with cylinders vs. highway MPG, estimating the relationship's significance through a trend line and correlation coefficient calculation.

Dashboard Creation:

1. Price Distribution by Brand and Body Style:

- Employed Excel's tools to create visualizations (histograms/box plots) showcasing car price distribution across brands and body styles.

2. Average MSRPs by Brand and Body Style:

- Utilized Excel to generate bar charts illustrating the highest and lowest average MSRPs, allowing for brand and body style comparisons.

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5. Brand-wise Variation in Horsepower, MPG, and Price:

- Excel scatter plots and grouped bar charts were utilized to depict variations in horsepower, MPG, and price across different car brands.

Through the Excel-centric approach, the project successfully delivered actionable insights for optimizing pricing, enhancing product development strategies, and maximizing profitability in the ever-evolving automotive industry. In conclusion, this project provided me with the practical exposure in using Formulas, Tables and Charts present in Excel.